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Launching Safely: Identifying Low-Risk Aircraft for Aviation Expansion

Overview

The company is entering the aviation sector, aiming to operate both commercial and private aircraft.

To ensure a safe and successful launch, we analyzed aircraft risk profiles across multiple dimensions.

This presentation outlines our findings and recommendations for selecting the safest aircraft to begin operations.

Why Aircraft Risk Matters

Aviation is a high-stakes industry where safety directly impacts brand trust, regulatory compliance, and operational costs.

Choosing low-risk aircraft reduces liability, improves insurance terms, and builds a strong foundation for growth.

Our goal: Identify aircraft models with the lowest historical risk to guide initial fleet acquisition.

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SOURCES

FAA aircraft incident reports, manufacturer safety records, operational data

SCOPE

100+ aircraft models across commercial and private categories

KEY VARIABLES

Investigation type (e.g., Accident vs. Incident), Aircraft damage level (e.g., Substantial, Destroyed, Minor), Injury severity (Fatal, Serious, Minor, Uninjured) and Aircraft make (manufacturer

DATA UNDERSTANDING

Data Analysis

How We Measured Risk

- Grouped data by investigation type and damage level to assess severity patterns
- Aggregated total injuries and uninjured counts to understand human impact
- Counted how often each aircraft make appeared under each damage category
- Identified the most frequently occurring aircraft make using statistical mode

DATA ANALYSIS

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KEY FINDINGS

- Most incidents were classified as “Accidents” with “Substantial” damage
- A small number of aircraft makes appeared disproportionately in high-damage categories
- Some makes consistently showed lower injury counts and fewer severe outcomes
- The most common aircraft make in the dataset was identified as a benchmark for comparison

AIRCRAFT SELECTION STRATEGY

Prioritize aircraft makes with:

- Low frequency of “Destroyed” or “Substantial” damage
- High proportion of “Uninjured” outcomes
- Fewer fatal or serious injuries per incident

Use this risk profile to guide procurement and fleet planning

Consider deeper analysis by model (not just make) in future phases

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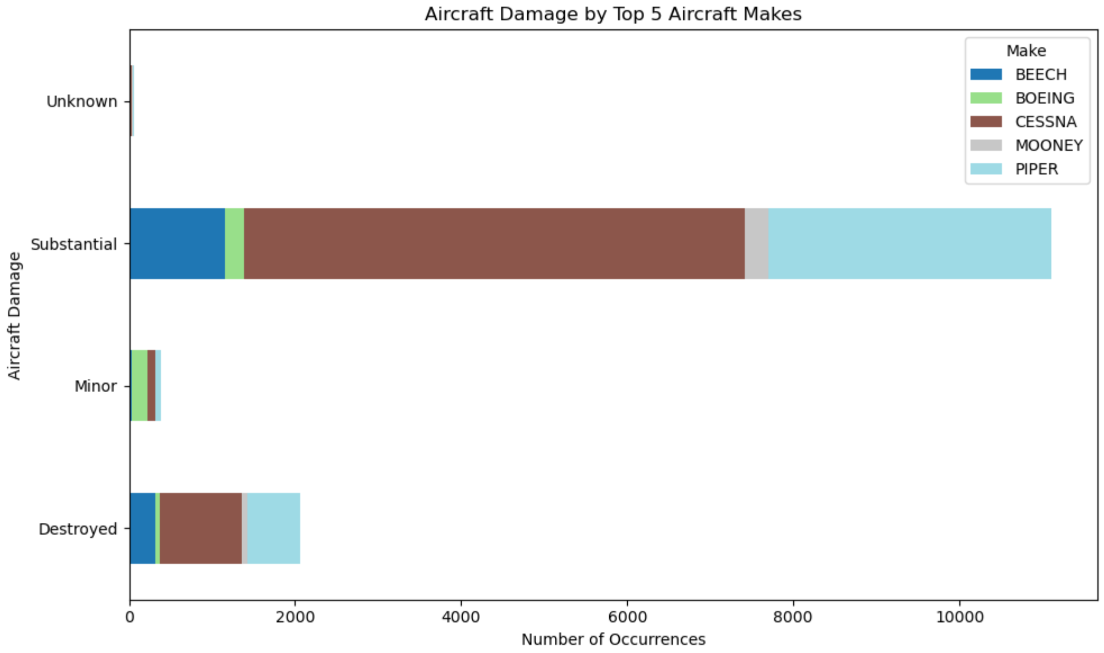

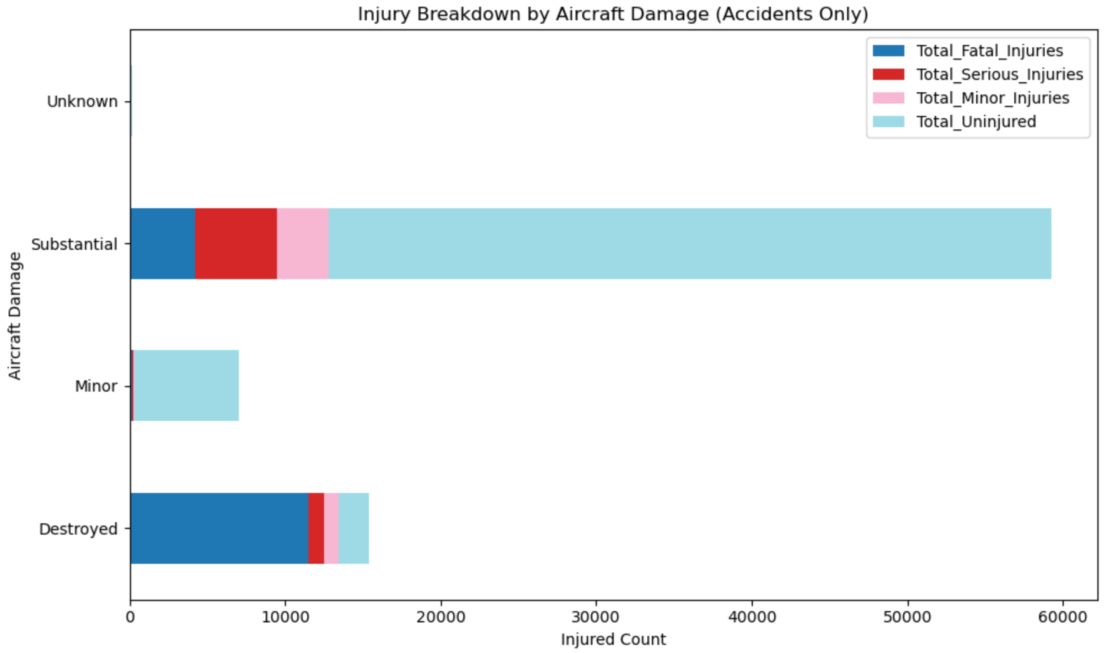

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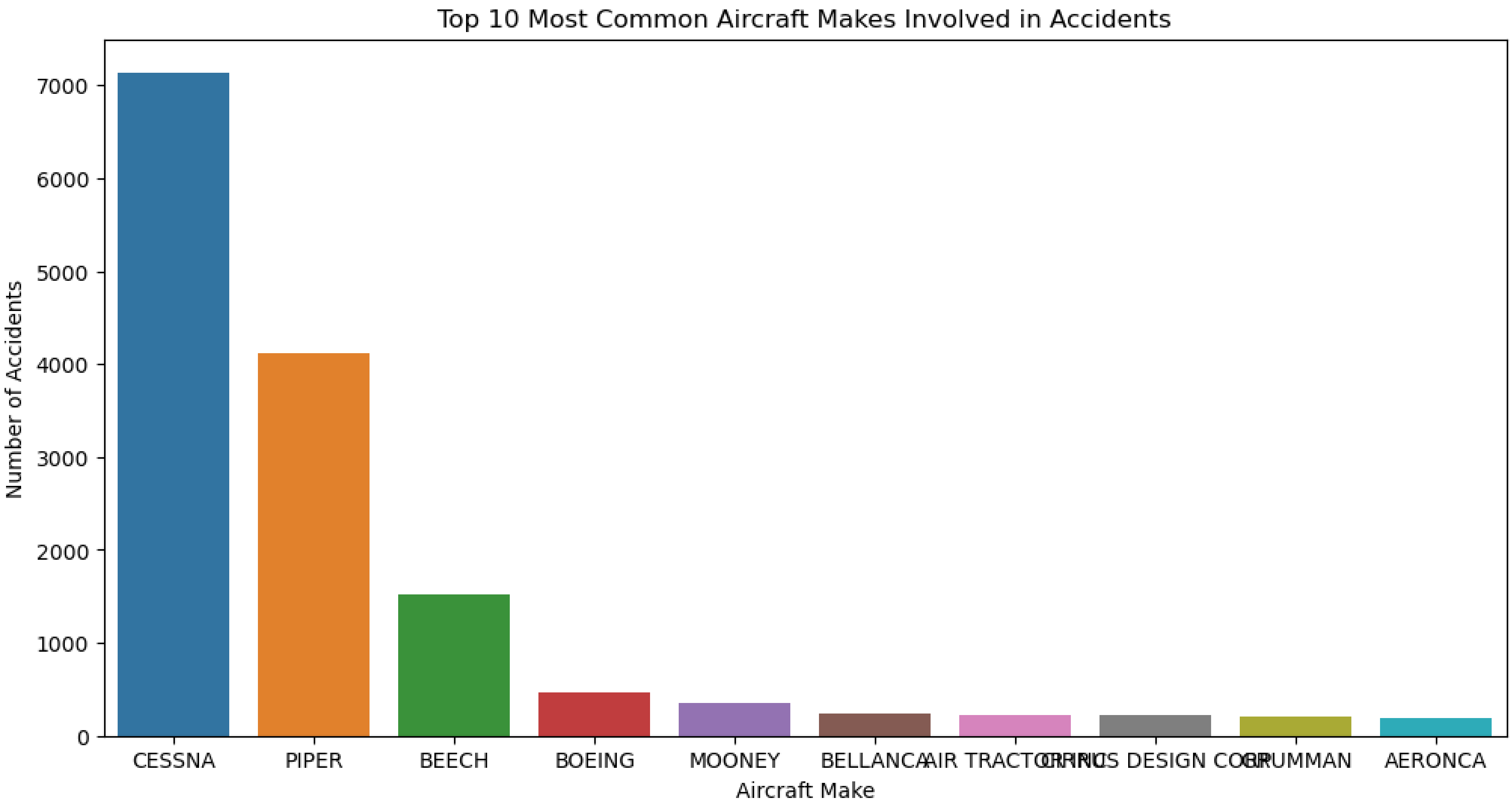

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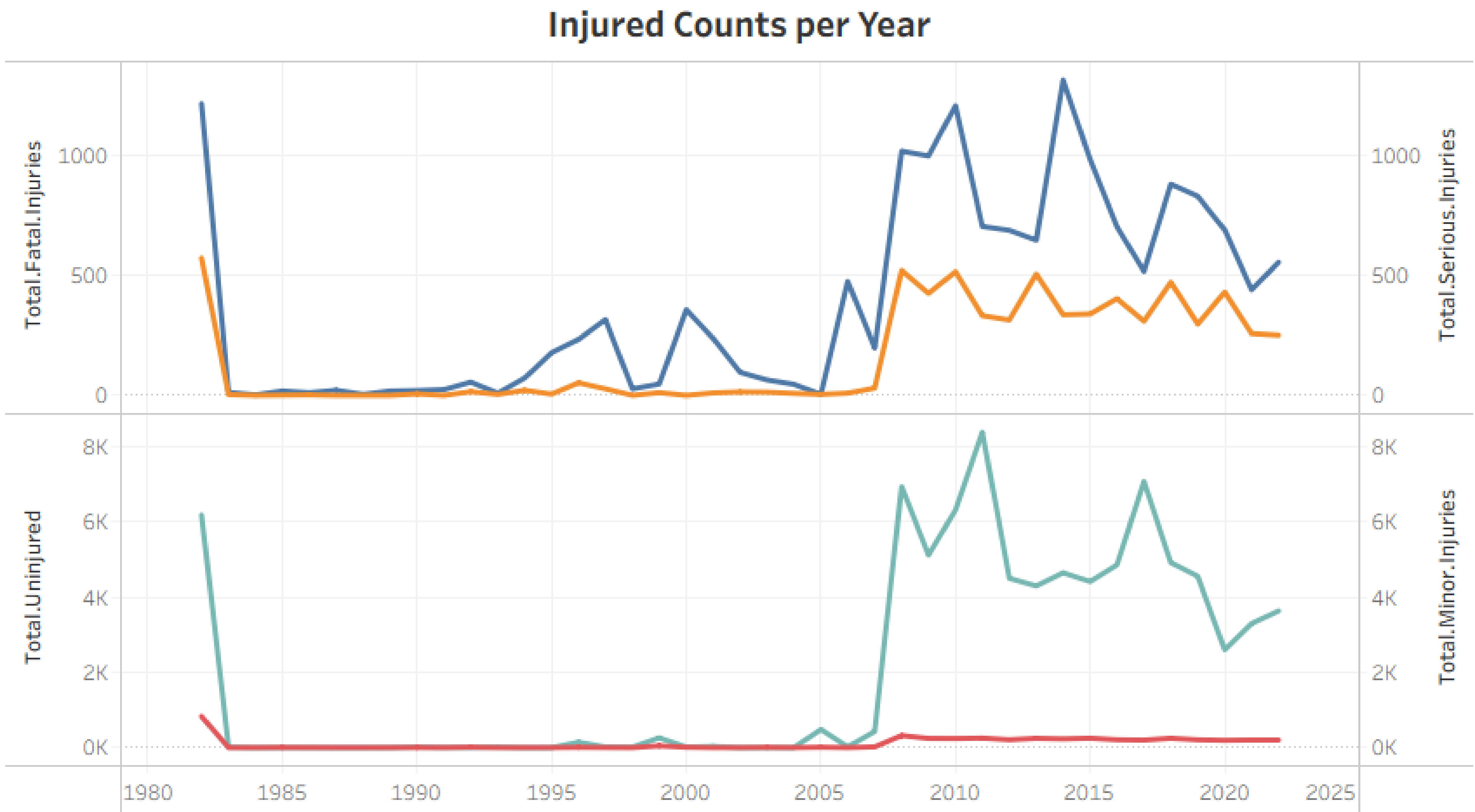
Injured Counts per Year
Total.Fatal.Injuries
Total.Serious.Injuries
Total.Uninjured
Total.Minor.Injuries
1000
500
0
8K
6K
4K
2K
0K
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025

FROM ANALYSIS TO ACTION

- Cross-reference findings with manufacturer support and maintenance costs
- Validate results with pilot and maintenance team feedback
- Build a dashboard to monitor ongoing aircraft safety trends
- Present shortlist of low-risk aircraft to aviation leadership for review

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THANK YOU

Ready for a Safe Takeoff?

We’re excited to support a safe and strategic entry into aviation

Any questions?

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